BSB
BURGUNDY SCHOOL OF BUSINESS
We believe in you

SESSION 2 OF 6 · 3 HOURS

Prompt Chains and Iterative Refinement

Building reliable, reusable AI outputs

MSc in AI and Digital Strategy Management
Advanced Prompting & API Integration



Session roadmap

Time	Block	Format
0:00	Recap and project task pitches	Plenary
0:20	From one prompt to a chain	Briefing + demo
0:45	Few-shot examples	Briefing
1:05	Workshop 1: build your chain	Individual, peer help
1:45	Break (10 min)	—
1:55	Test, score and refine	Briefing
2:20	Workshop 2: test and refine	Pairs
2:50	Submit deliverable, Session 3 preview	Plenary

From one prompt to a chain

Example task: handling a customer complaint email

1 Extract

Pull the key facts from the customer email into a fixed JSON format

→

2 Classify

Tag topic and urgency, guided by three labelled examples

→

3 Draft

Write a reply in brand tone using the tags

→

4 Check

Review the draft against the quality checklist

One step, one job, one input, one output format.

Few-shot examples

ZERO-SHOT

Instruction only: “Classify this email as billing, delivery or product.”

Result: labels and wording vary between runs

FEW-SHOT

Instruction + 2–3 worked examples (email → label) + the new email

Result: the model copies the pattern and the format

Choose examples that cover edge cases · keep the format identical · 2 to 5 examples are usually enough.

WORKSHOP 1 · 40 MIN · INDIVIDUAL, PEER HELP

Build your chain

1. Split your task into 3–4 steps
2. Write one prompt per step with a fixed output format
3. Add 2 examples where output varies
4. Run it on one real input

Output: chain v1 with a prompt and format for every step



Test, score, refine

Test	Score	Diagnose	Refine

Quality criterion	Example check
Accuracy	Facts match the source text
Format	Template or JSON followed exactly
Tone	Matches the brand voice
Consistency	Same quality on 5 varied inputs

Log every run: version · input · output · score · change made.

WORKSHOP 2 · 30 MIN · PAIRS

Test and refine

1. Run the chain on 3 varied inputs
2. Score each run on your criteria
3. Change one thing, then re-run the same 3
4. Log every version in the test log

Output: test log with at least two versions



Try to break it with

- A very short input
- A long, messy input
- A different tone or language
- Missing information

Deliverable and next session

SUBMIT BEFORE LUNCH

Draft prompt chain with documented test results

Chain steps · 3 test inputs · quality criteria · test log with 2 versions

Email one PDF or Word file named Surname_Session2_Chain to fabien.gueret-jullien@bsb-education.com

Subject: [Session 2] Surname – Chain draft. No personal or confidential data in your test inputs.

SESSION 3 · AFTER LUNCH

API Fundamentals for Non-Engineers

Over lunch: check your AI account access (10 min). We read code and use low-code tools; no programming is required.